Offensive Play Style Clustering and EPA per Drive Prediction in the NFL





Executive Summary

What this project accomplished:

- Built 2018–2024 NFL team-season offensive profiles
- Identified 4 distinct offensive style clusters
- Modeled EPA (Expected points added) per Drive using Ridge Regression ($R^2 \approx 0.84$)
- Analyzed cluster stability, next-season predictability, and playoff appearance %

Main takeaway:

Offensive style clusters explain efficiency within-season, but do not reliably predict future efficiency or playoff qualification alone.

Problem Statement

Goal
Identify offensive play-style “archetypes” and determine how they relate to efficiency (EPA per drive) and team success.

Why this matters

- EPA reflects true scoring value
- Play-style identity influences coaching, roster building
- Strategic patterns can reveal competitive advantages

Dataset Overview

Data Sources

- nflfastR play-by-play datasets (2018–2024)
- ~400k plays, aggregated into 224 team-season profiles

Tools Used

- Python (pandas, NumPy)
- scikit-learn (KMeans, Ridge, StandardScaler)
- seaborn / matplotlib
- Jupyter Notebook

Data Levels Used:

- Play → Drive → Team → Season

Feature Engineering

Key variables constructed:

- pass_rate, early_down_pass_rate
- success_rate
- epa_per_play
- avg_air_yards, avg_yac
- shotgun_rate, no_huddle_rate
- epa_per_drive (target)

```
(   season posteam  plays  pass_rate  early_down_pass_rate  success_rate  \
 0    2018      ARI    896   0.611607              0.546392      0.345982
 1    2018      ATL   1008   0.658730              0.603073      0.462302
 2    2018      BAL   1116   0.525986              0.485516      0.452509
 3    2018      BUF   1001   0.541459              0.473958      0.376623
 4    2018      CAR    998   0.596192              0.565107      0.477956

    epa_per_play  fourth_down_go_rate  avg_air_yards   avg_yac  shotgun_rate  \
 0     -0.254928             0.017857       8.054656 -1.222098      0.574777
 1      0.071760             0.018849       8.555195 -0.318452      0.510913
 2      0.020948             0.019713       8.685353 -0.980287      0.801075
 3     -0.149135             0.016983       9.925703 -1.889111      0.572428
 4      0.055432             0.019038       7.443649 -0.090180      0.767535

    no_huddle_rate  epa_per_drive
 0        0.071429      -1.260814
 1        0.073413       0.435749
 2        0.060932       0.139003
 3        0.074925      -0.781591
 4        0.081162       0.333260  ,
 (224, 13))
```

Preprocessing Summary

Actions completed:

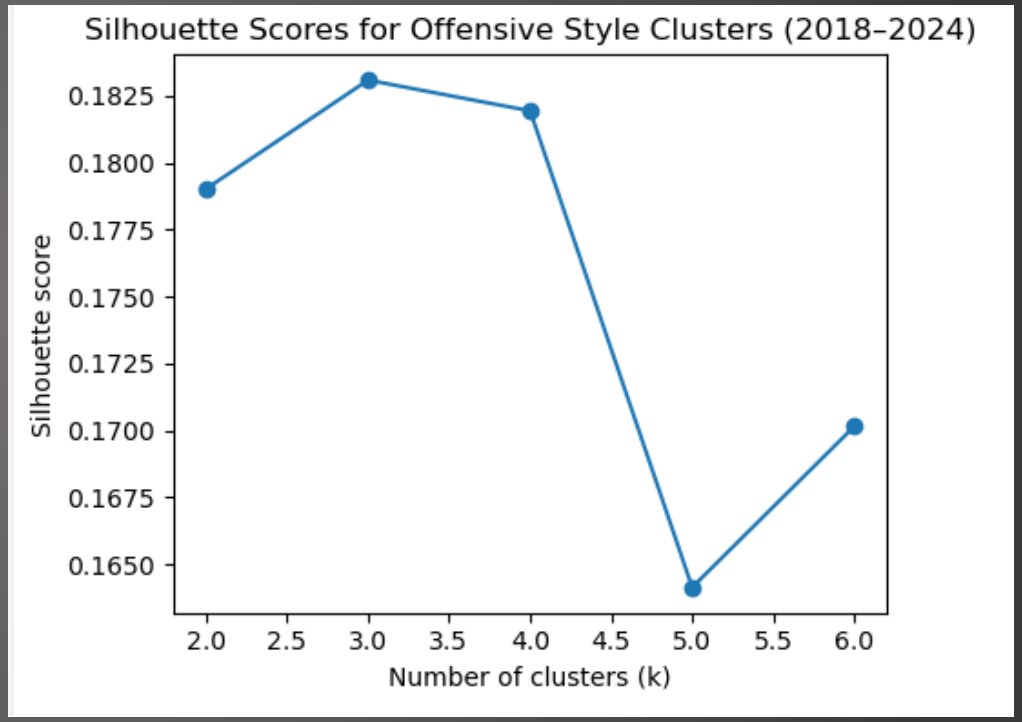
- Filtered out penalties, timeouts, missing EPA plays
- Calculated per-drive summaries
- Aggregated drives → team-seasons
- Standardized features for clustering
- Normalized distributions (z-score scaling)
- Removed extremely rare edge cases
- Final feature matrix: **224 × 9**

Outcome: Clean team-season matrix with 9 standardized features.

Clustering Approach

Method: K-Means
Range tested: k=2–6
Metric: Silhouette score

Best ~0.183 at k=3, but k=4 chosen for better interpretability.

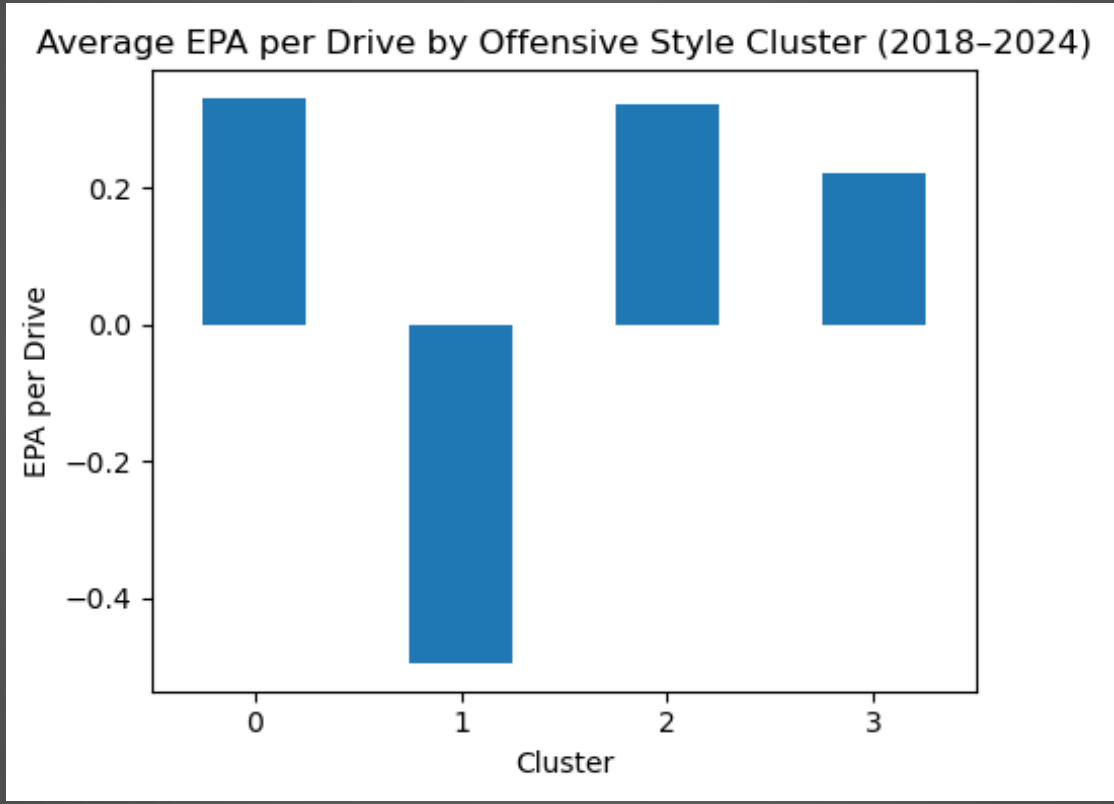


The Four Offensive Clusters

Cluster 0 - Pass-Heavy Efficient
Cluster 1 - Inefficient Balanced / Struggling
Cluster 2 - Efficient Balanced
Cluster 3 - Vertical / Aggressive

cluster	pass_rate	early_down_pass_rate	success_rate	epa_per_play	fourth_down_go_rate	avg_air_yards	avg_yac	shotgun_rate	no_huddle_rate	epa_per_drive
0	0.630	0.586	0.460	0.053	0.018	7.831	-0.227	0.702	0.092	0.331
1	0.604	0.549	0.405	-0.089	0.024	7.964	-0.745	0.681	0.099	-0.495
2	0.555	0.504	0.456	0.053	0.019	7.517	0.071	0.554	0.073	0.324
3	0.525	0.477	0.448	0.035	0.020	8.555	-0.559	0.775	0.128	0.221

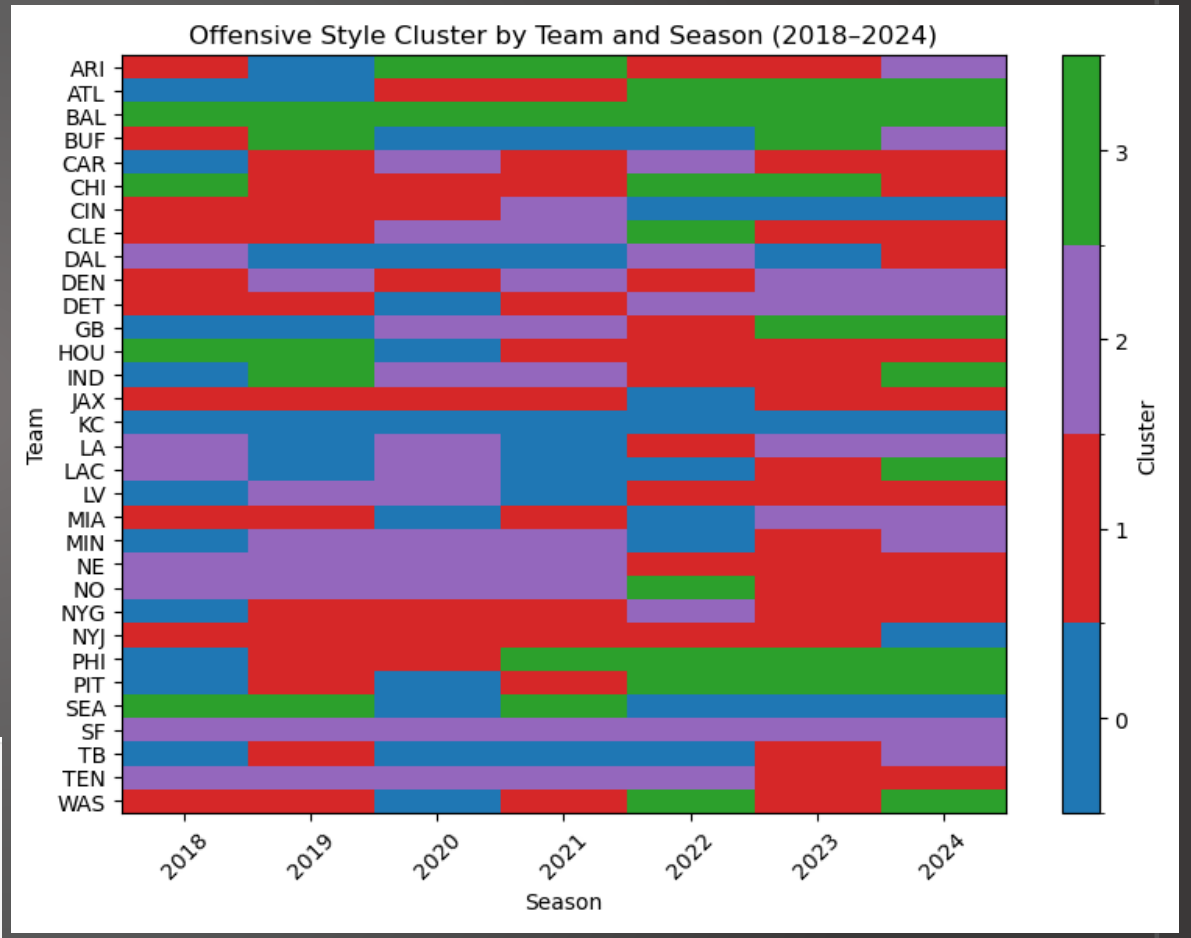
Cluster EPA Comparison



Cluster 1 is clearly below zero → validates interpretability.

Identity Stability (2018–2024)

- ~40% year-to-year cluster stability
- Style changes often follow QB/HC changes
- Clusters 0, 2, and 3 have remained stable with at least one team
- Cluster 1 has not remained stable on any team (Volatile)




```
# Add stability analysis

# Measure how often teams stay in the same cluster:

team_shift['cluster_stable'] = (
    team_shift['cluster'] == team_shift.groupby('posteam')['cluster'].shift(1)
)
team_shift['cluster_stable'].mean()
```

```
0.4010416666666667
```

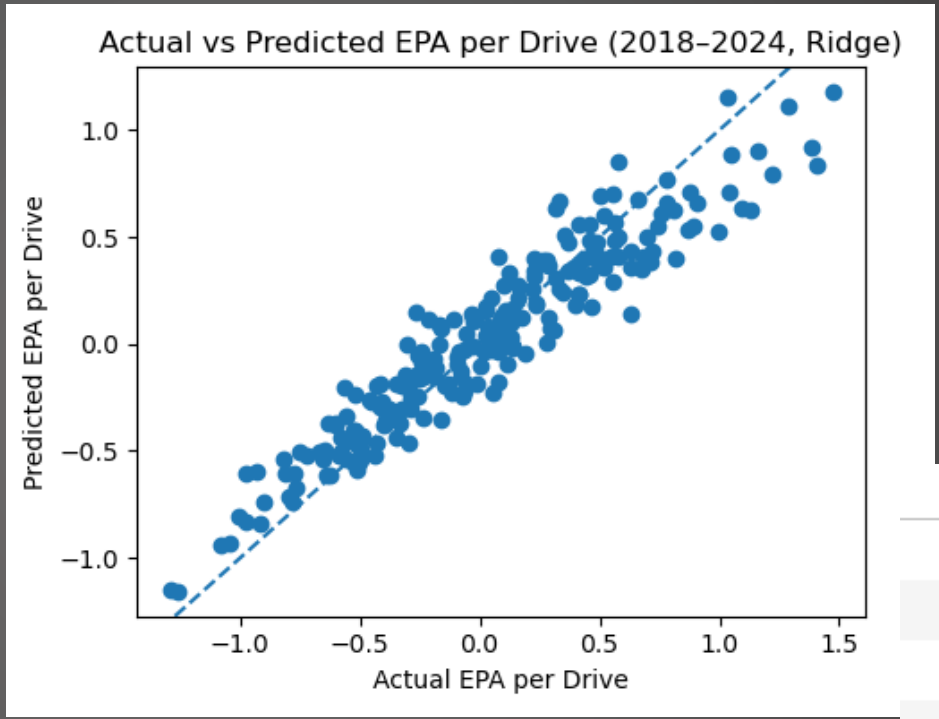
Regression Model (EPA per Drive)

Results (7-fold CV):
- $R^2 \approx 0.84$
- MAE ≈ 0.16

Top predictors:

- EPA per play
- Success rate
- YAC
- Air yards

Main takeaway: Within-season efficiency is highly predictable.



feature	coefficient
epa_per_play	2.430449
success_rate	0.778381
avg_yac	0.601928
avg_air_yards	0.288446
shotgun_rate	0.257389
early_down_pass_rate	0.093462
fourth_down_go_rate	-0.023040
pass_rate	-0.043081
no_huddle_rate	-0.051885

Predicting Next-Season EPA

Goal: Evaluate whether clusters can predict future efficiency.

Method:

- Shift EPA per drive by +1 season
- Use cluster labels (one-hot encoded) to predict next-season EPA

Preliminary Results:

- $R^2 \approx$ **0.07** → weak predictive power
- MAE ≈ **0.41** → moderately large error

Interpretation:

- **Cluster identity alone doesn't predict next-season EPA**
- Supports idea that roster/coaching turnover drives future efficiency more than style cluster alone

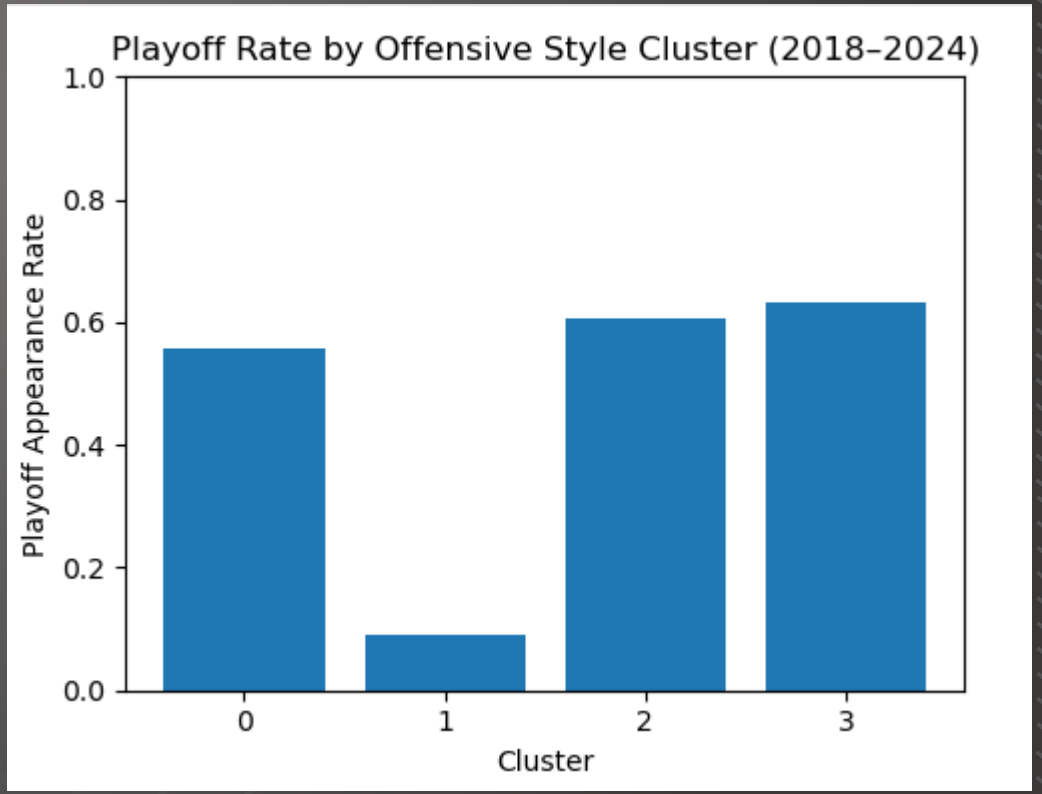
Playoff Appearance Analysis

Process:

- Extract POST season games
- Build list of playoff team-seasons
- Merge with cluster labels
- Compute playoff rate per cluster

Main finding: Efficient clusters reach playoffs far more often

	cluster	playoff
0	0	0.557692
1	1	0.089744
2	2	0.607143
3	3	0.631579



Key Insights

Synthesis slide:

- Four distinct offensive archetypes exist
- EPA per drive is driven mostly by EPA/play, success rate and average yards after catch
- Styles change frequently year-to-year
- Teams in clusters 0, 2 and 3 reached the playoffs more often than average, suggesting this style is associated with successful seasons
- This suggests that moving away from a Cluster-1 style and toward the other clusters may increase a team's chances of a 'successful' season, though offense alone doesn't guarantee playoff qualification.

Limitations and Considerations

- Offense is only part of team success
- Injuries, QB/coaching changes could impact a season
- Cluster styles don’t imply causation
- EPA/drive varies with strength of schedule

Challenges

Challenges:

- Large dataset (400k+ plays) required careful memory management
- Aggregation was complex (plays → drives → teams → seasons)
- Season-to-season team identity is unstable, reducing predictive power

Adjustments:

- Expanded feature set for clustering
- Tested multiple k-values
- Added next-season EPA prediction + cluster-stability analysis
- Added playoff qualification as a new evaluation dimension

What I learned

- Building drive-level and team-season features is complex but powerful
- Clustering is interpretive, domain knowledge matters
- EPA per play is the backbone of modern offensive analysis
- Predicting NFL future performance is noisy due to personnel turnover
- Real data science involves iteration, testing, and reframing the question

Conclusion

This project shows:

- Play-style clusters offer meaningful insight into team identity
- Efficiency is highly predictable within-season
- But future performance hinges on roster/coaching changes more than style

Future work:

- Add defensive features
- Incorporate roster continuity metrics
- Test tree-based or ensemble models for prediction



Thank you

